

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
7			1. Neutron decays into a proton, electron and an electron antineutrino 2. Charge conservation shown: $0 \rightarrow +1 + (-1) + 0$ 3. Baryon conservation shown: $1 \rightarrow 1 + 0 + 0$ or quark number conservation shown: $3 \rightarrow 3 + 0 + 0$ 4. Lepton conservation shown: $0 \rightarrow 0 + 1 + (-1)$ 5. Quark analysis shown: $udd \rightarrow uud$ or d quark ‘becomes’ an u quark so neither u quark number or d quark number are individually conserved there is a change in quark flavour 6. This is a ‘weak’ interaction (or interaction involving the weak force) because neutrino involvement or change in quark flavour 5-6 marks 5 or 6 of the above areas covered accurately <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured.</i> 3-4 marks 3 or 4 of the above areas covered accurately or attempt made at 5 or 6 areas <i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure.</i> 1-2 marks 1 or 2 of the above areas covered accurately or attempt made at 3 or more areas <i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure.</i> 0 marks <i>No attempt made or no response worthy of credit.</i>	4	2		6		
			Question 7 total	4	2	0	6	0	0